Toulouse School of Economics, 2013-2014
Macroeconomics L3 Semester 1 – Franck Portier

Problem set 5: (very) Long Run growth (part 2)
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Problem I – Children as a inferior good

To explain the demographic transition, economists need to account for a decrease in the number of children per woman when income per capita increases. One simple way to do it is to assume that people value the number of children they have n, and that n enters the utility function. In this problem, we consider various utility functions and check whether n is an inferior good (a good whose demande decreases when income increases) or not. Let us consider a static problem where utility is

$$u(m,n)$$

and where m represents the consumption of manufacturing goods and n the number of children. We will always assume increasing and concave utility functions. The price of manufacturing goods is normalized to 1 and the price of children is p (in class, we assumed that children are costly because they need food, and that the price of food is p). W is the income of the household so that the budget constraint is

$$m + pn = W.$$

1 – Compute the optimal demand for children when $u = \log m + \gamma \log n$. Compute the sign of $\frac{\partial n}{\partial W}$. Discuss.

2 – Compute the optimal demand for children when $u = m + \gamma \log n$. Compute the sign of $\frac{\partial n}{\partial W}$. Discuss.

3 – Compute the optimal demand for children when $u = \log(m + \gamma \log n)$. Compute the sign of $\frac{\partial n}{\partial W}$. Discuss.

It is not easy to find simple utility functions for which n is an inferior good (at least in some range). In the following, we will study the case $u = \min(v, \widetilde{v})$ with $v = m + B$ and $\widetilde{v} = A(m+n)$. We assume $A > 1$ and $B > 0$.

4 – Compute the equation of the indifference curve of level $\overline{u}$ for the three cases $\overline{u} < B$ (region I), $B < \overline{u} < \frac{AB}{A-1}$ (region II) and $\overline{u} > \frac{AB}{A-1}$ (region III). Draw them in the plane (m,n). Show that in region II, indifference curves have a kink and that this kink is on the line $n = \frac{B-(A-1)m}{A}$.

5 – Show graphically that if "p is small enough" and "W is neither too small nor too large", then the optimal solution of the maximization problem will be at a kink. Sow that in that case n is an inferior good. (this corresponds to region II)

6 – Derive analytically the conditions "p is small enough" and "W is neither too small nor too large" under which the optimal solution of the maximization problem is at a kink.

7 – Give the expression of optimal n in that case, and show that $\frac{\partial n}{\partial W} < 0$.

Problem II – A model of unified growth in which adults consume food

We have studied in section 6 of Lecture 5 a model of unified growth. One unrealistic model assumption was that adults do not consume food, but only buy food for there children. In this problem, we remove this assumption and compute again the model equilibrium.

Adults like manufacturing goods m_t, adult food f_t and children n_t. Their utility is given by

$$u_t = m_t + \gamma \log n_t + \delta \log f_t.$$

As in the lecture, the manufacturing good is the numéra ire and one children costs one unit of food, so that the budget constraint of an adult in t is :

$$p_t n_t + p_t f_t + m_t = w_t.$$

1 – Solve the utility maximization problem of an adult and derive the demand for children and the demand for agricultural goods (food for chidden and for adults).

2 – Show that total demand for food is $\frac{(\gamma+\delta)}{p_t} L_t$.

As in the lecture, technology is for food $Y_t^A = A_t^{\epsilon}(L_t^A)^{\alpha} X^{1-\alpha}$, where X is land, assumed to be in fixed supply, and normalized to $X = 1$; L^A is agricultural labor and A is productivity on agriculture. ϵ and α are between 0 and 1. The technology for the manufacturing good is $Y_t^M = M_t^\epsilon L_t^M$. Note that we assume for simplicity that land is free disposal (land rent is zero). We finally assume Learning-By-Doing, so that $A_{t+1} - A_t = Y_t^A$ and $M_{t+1} - M_t = Y_t^M$.

3 – Write the equilibrium condition on the food market and derive from that condition an expression for the share of agricultural labor in total labor $\theta_t = \frac{L_t^A}{L_t}$. How is this expression different from the one in the lecture? Discuss.

4 – Give an expression for the wages in the two sectors.

5 – Why free movements of workers between sectors will equalize the wages? From the wage equality condition, derive an expression for p_t as a function of M_t, A_t, θ_t and L_t.

6 – Using the demand for children, derive n_t as a function of M_t, A_t, and L_t. How is this expression different from the one in the lecture? Discuss.

7 – Show that the long run level of population growth is the same than in the model of Lecture 5.

Problem III – Simulation of the model of unified growth

Take the model of the last part of section 6 of Lecture 5 with $Y_t^A = \mu A_t^\epsilon (L_t^A)^\alpha X^{1-\alpha}$ and $Y_t^M = \delta M_t^\phi L_t^M$. The model solution is $n_t = \mu \left(\frac{\gamma}{\delta}\right)^\alpha \frac{A_t^\epsilon}{M_t^{\alpha\phi}(\gamma L_t)^{1-\alpha}}$ and $\theta_t = \left(\frac{n_t L_t^{1-\alpha}}{\mu A_t^\epsilon}\right)^{\frac{1}{\alpha}}$.

Assume that parameters are $\alpha = 0.8$, $\epsilon = 0.45$, $\phi = 0.3$, $\mu = 0.5$, $\delta = 1.5$, $\gamma = 3.4$ and initial values are $A_0 = 0.8$, $M_0 = 18.15$ and $L_0 = 0.014$. It is assumed that the length of the period is 25 years (one generation), so that growth factors needs to be taken at the power $1/25$ to be expressed in annual terms.

Use a spreadsheet to reproduce the figures for time evolution of g_L, g_A, g_M and w. Print the figures and the spreadsheet with the computation.